*CS319:*

# Internet Programming I

*Introduction to Internet Programming*



BANGKOK UNIVERSITY
THE CREATIVE UNIVERSITY

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# Why Learn Programming in the Era of AI??

*“Coding is not dead. AI just raises the level of abstraction. You still need to understand what is happening under the hood to build, debug, and steer complex systems.”*

**AI is an Amplifier, Not a Replacement.**

“A person who understand coding and using AI
And another person who just using AI”

***Who is better?***

MODULE #1

# *Front-End Web Development*

*Web Programming*

# Agenda

1. Introduction to front-end web development
2. HTML 5
3. CSS – Cascading Style Sheet
4. JavaScript



# *What is World Wide Web?*

## (Web)

# Who invented Web?

**Sir Tim Berners-Lee and his colleague Robert Cailliau.**

- Developed while working CERN (European Organization for Nuclear Research) in Switzerland.

**Why they made it?**

- To solve the challenge of sharing research data among scientists worldwide collaborating at CERN.
- To connect documents across different computers using hyperlinks, allowing users to navigate between related information over the internet.

# Who invented Web? (Continue)

**Three Core Technologies Created by Tim Berners-Lee:**

- **HTML (HyperText Markup Language):** The formatting language used to structure and present web content.
- **HTTP (HyperText Transfer Protocol):** The standard communication protocol used to transmit data between browsers and web servers.
- **URL (Uniform Resource Locator):** The global addressing system used to identify and locate resources on the internet.

**Web/HTML = Document**

*(Word and Google docs is a document too!)*

# Website?

**Definition:** A collection of interconnected web pages (digital documents), multimedia, and digital assets published under a single domain name, retrieved and displayed to users through a web browser using standard protocols (HTTP/HTTPS).

**Core Purpose:** To deliver information, interactive services, and user experiences across the Internet.

# How to develop a website?

## Frontend Development (Client-Side):

- Implement client-side structure, styling, and interactivity using HTML, CSS, and JavaScript.
- Build responsive, accessible user interfaces using modern frameworks/libraries (e.g., React, Vue).

## Backend Development (Server-Side):

- Implement business logic, authentication, and RESTful APIs using languages like Node.js, Python, or PHP.
- Design and manage database systems (e.g., PostgreSQL, MongoDB) for data persistence.

# Type of Web Document

**1. Static:** A static web document resides in a file that it is associated with a web server. The author of a static document determines the contents at the time the document is written. Because the contents do not change, each request for a static document results in exactly the same response.

**2. Dynamic:** A dynamic web document does not exist in a predefined form. When a request arrives the web server runs an application program that creates the document. The server returns the output of the program as a response to the browser that requested the document. Because a fresh document is created for each request, the contents of a dynamic document can vary from one request to another. Such as PHP, ASP, ASP.NET, JSP



**3. Active:** An active web document consists of a computer program that the server sends to the browser and that the browser must run locally. When it runs, the active document program can interact with the user and change the display continuously. Such as JAVA APPLET, ACTIVE X, VB SCRIPT or JavaScript, WebAssembly

# Pros and cons of each document type

## 1. Static Web Documents

### Pros

- **High Performance & Fast Delivery:** Pre-rendered files are served instantly with minimal latency.
- **Low Server Overhead:** Requires very little CPU and RAM since no script execution is needed on the server.
- **Easy Caching & Scalability:** Highly compatible with CDNs (Content Delivery Networks) to handle high traffic effortlessly.
- **Stronger Security:** No backend databases or server-side scripts to exploit via injection attacks.

### Cons

- **No Personalization:** Serves the exact same content to every user.
- **High Maintenance Effort:** Changing shared elements (e.g., navigation bar) requires updating every individual file manually unless automated by a build tool.
- **No Real-Time Capabilities:** Cannot handle user input, transactions, or live data updates directly.

# Static Web Documents Example




*Portfolio or Resume website*

# Pros and cons of each document type

## 2. Dynamic Web Documents

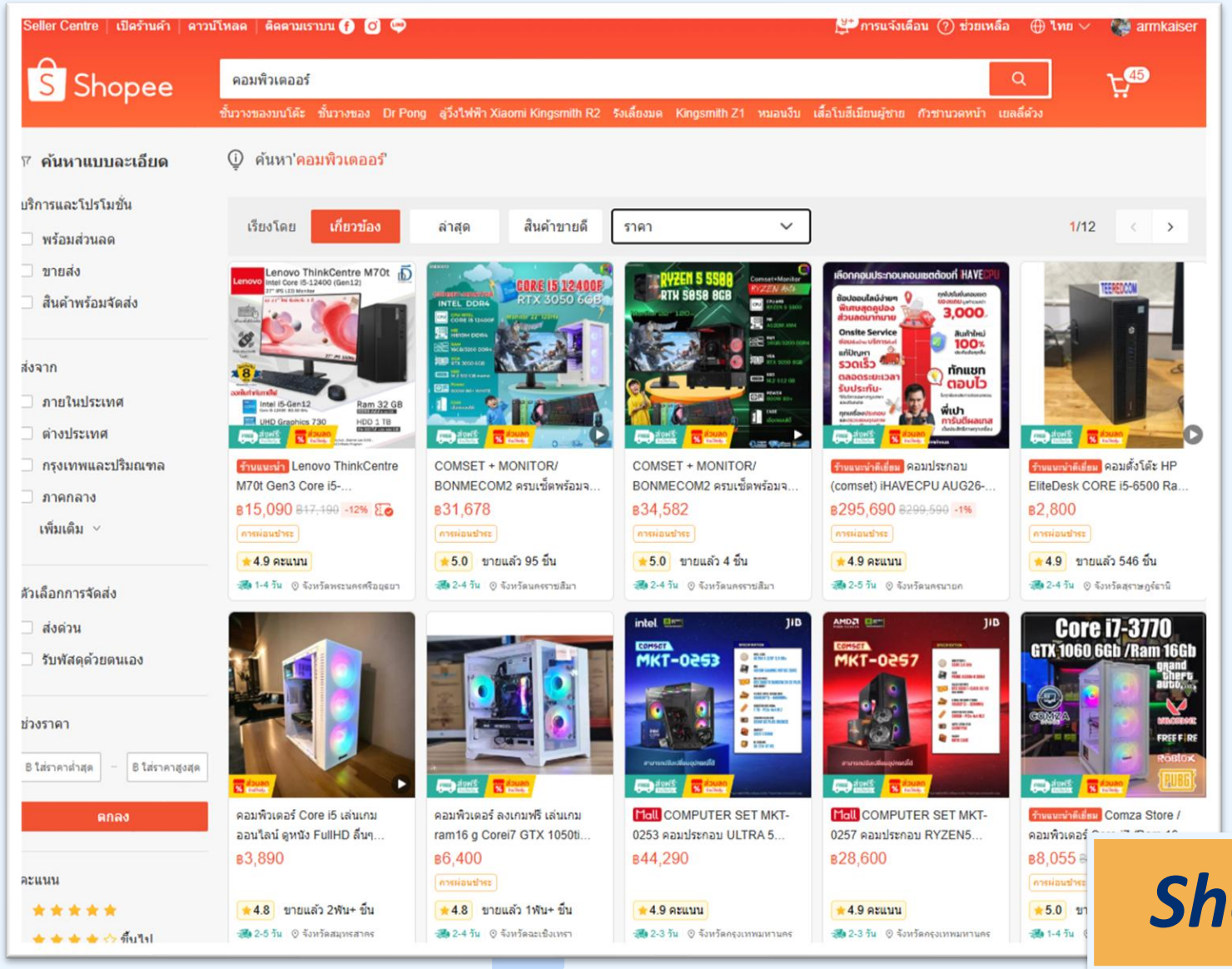
### Pros

- **Personalized & Context-Aware:** Generates tailored content based on user identity, session state, content, location, or time.
- **Centralized Data Management:** Connected to databases, allowing a single template to render thousands of distinct pages.
- **Easy Content Updates:** Content creators can update databases or CMS platforms without modifying underlying HTML files..

### Cons

- **Server Processing Latency:** Increased Time to First Byte (TTFB) due to server-side script execution and database queries.
- **Higher Infrastructure Costs:** Demands greater server computational power and database scaling under heavy traffic.
- **Larger Attack Surface:** Vulnerable to server-side exploits such as SQL Injection (SQLi) and Remote Code Execution (RCE).

# Dynamic Web Documents Example



# Pros and cons of each document type

## 3. Active Web Documents

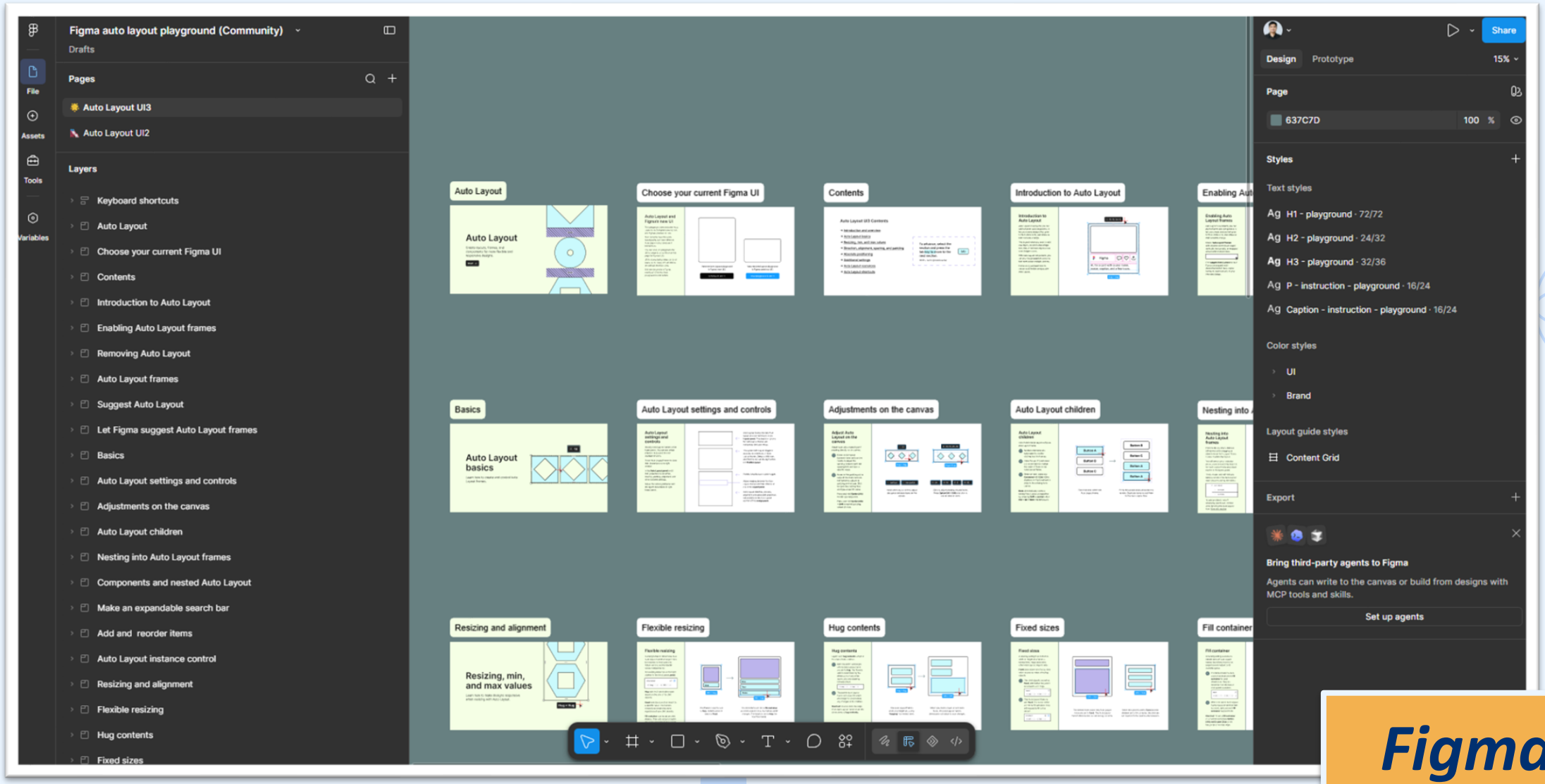
### Pros

- **Rich Interactivity:** Delivers immediate feedback and desktop-like user experiences without requiring full-page reloads.
- **Offloaded Computation:** Shifts execution workload from the central server to individual client machines.
- **Reduced Network Bandwidth:** Transmits raw data payloads (e.g., JSON via APIs) rather than re-sending entire HTML documents on every interaction.

### Cons

- **Client Hardware & Browser Dependency:** Performance varies widely depending on the end-user's device capabilities and browser engine compatibility.
- **Initial Load Overhead:** Requires downloading and parsing large script bundles before the application becomes fully interactive.
- **Client-Side Security Risks:** Code is exposed to inspection and vulnerable to client-side attacks such as Cross-Site Scripting (XSS).

# Active Web Documents Example




Figma

# Website VS Web app

## Website

- **Primary Goal:** Focuses on content consumption (reading, browsing, and gathering information).
- **User Interaction:** Simple, one-way navigation (clicking links, scrolling, basic search).
- **Authentication:** Mostly public; login is optional or used only for simple features.
- **Tech Stack:** HTML/CSS, Content Management Systems (CMS like WordPress), Server-Side Rendering.
- **Data Flow:** Classic page reload (Client requests → Server returns whole new HTML page).
- **Examples:** News portals, corporate pages, blogs (e.g., Wikipedia, BBC).

## Web App

- **Primary Goal:** Focuses on task execution (creating, editing, and manipulating data).
- **User Interaction:** Rich, two-way interaction with real-time updates (no full-page reloads).
- **Authentication:** Essential; login defines user rights, personal data, and security rules.
- **Tech Stack:** Modern Frontend Frameworks (React, Vue), REST/GraphQL APIs, Databases.
- **Data Flow:** Asynchronous API calls (JSON data updates specific UI components without page reloads).
- **Examples:** Cloud tools, interactive software (e.g., Google Docs, Figma, Trello).

# Web Browser

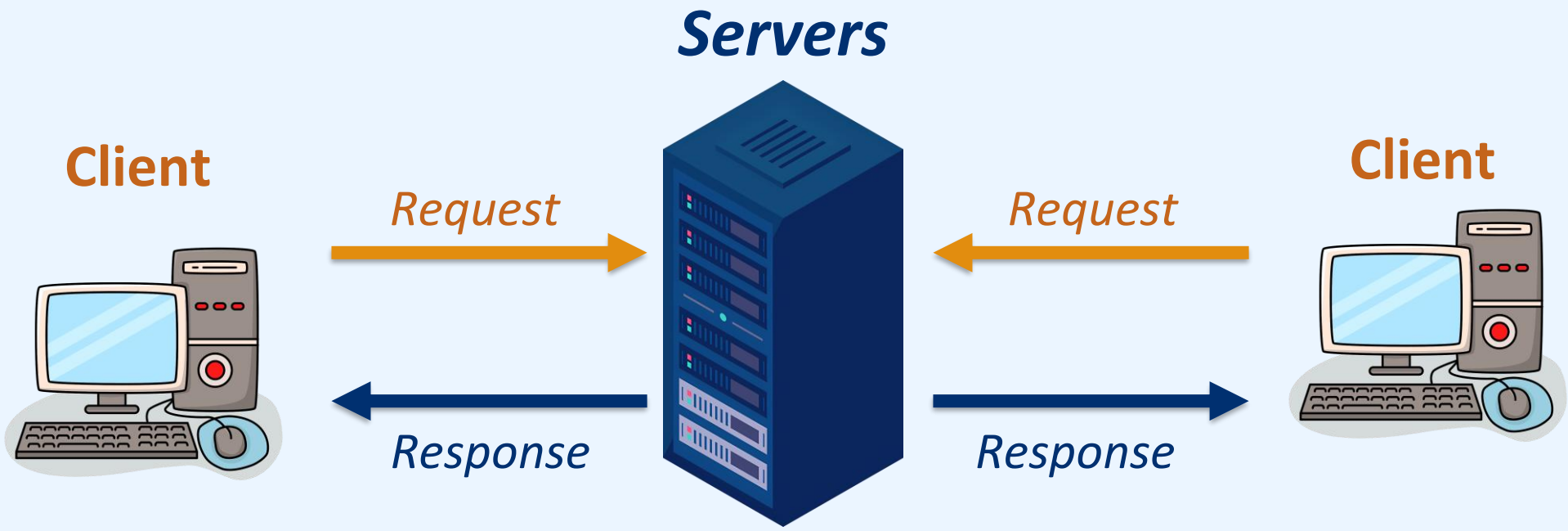
**Definition:** A client-side software application designed to retrieve, interpret, and render content from the World Wide Web.

**Components:**

1. **Rendering Engine:** Parses HTML/CSS and draws the pixels on screen
2. **JavaScript Engine:** Compiles and executes client-side JavaScript code
3. **Networking Layer:** Handles communication, HTTP requests, DNS lookup, and caching
4. **Security & Storage:** Manages SSL/TLS encryption, cookies, local storage, and isolates pages via the Same-Origin Policy.



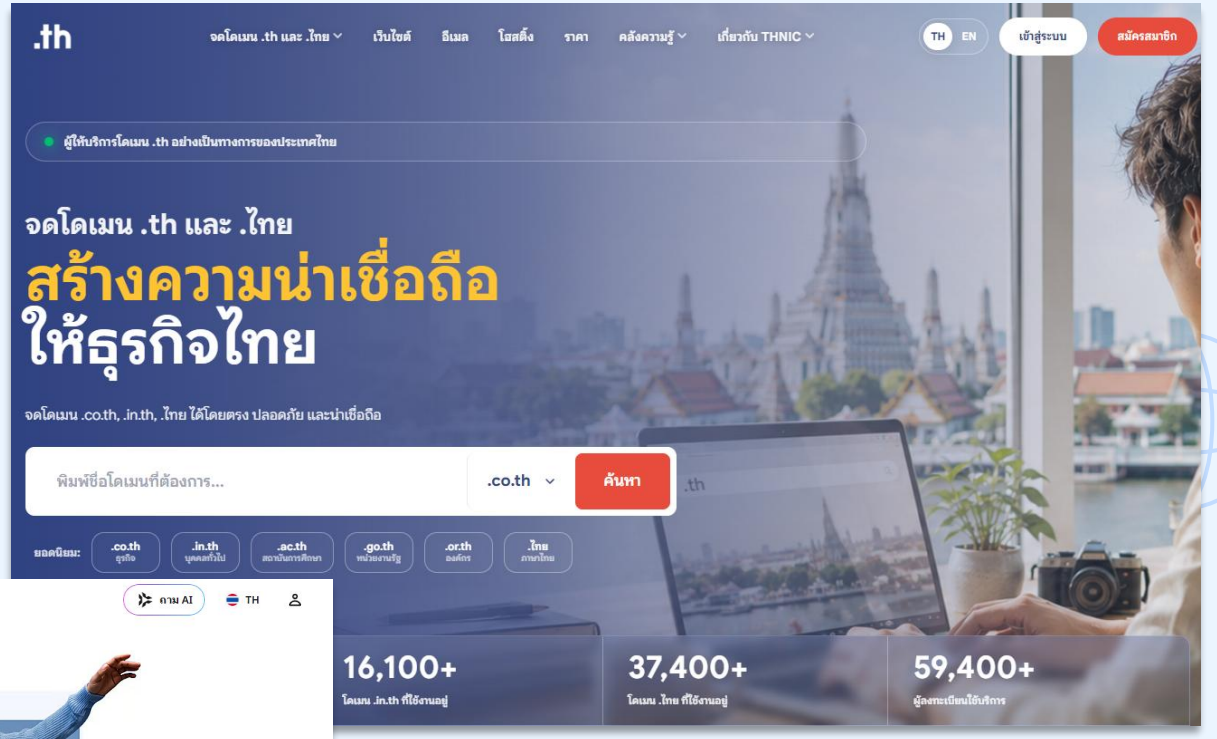
# Client Server Architecture



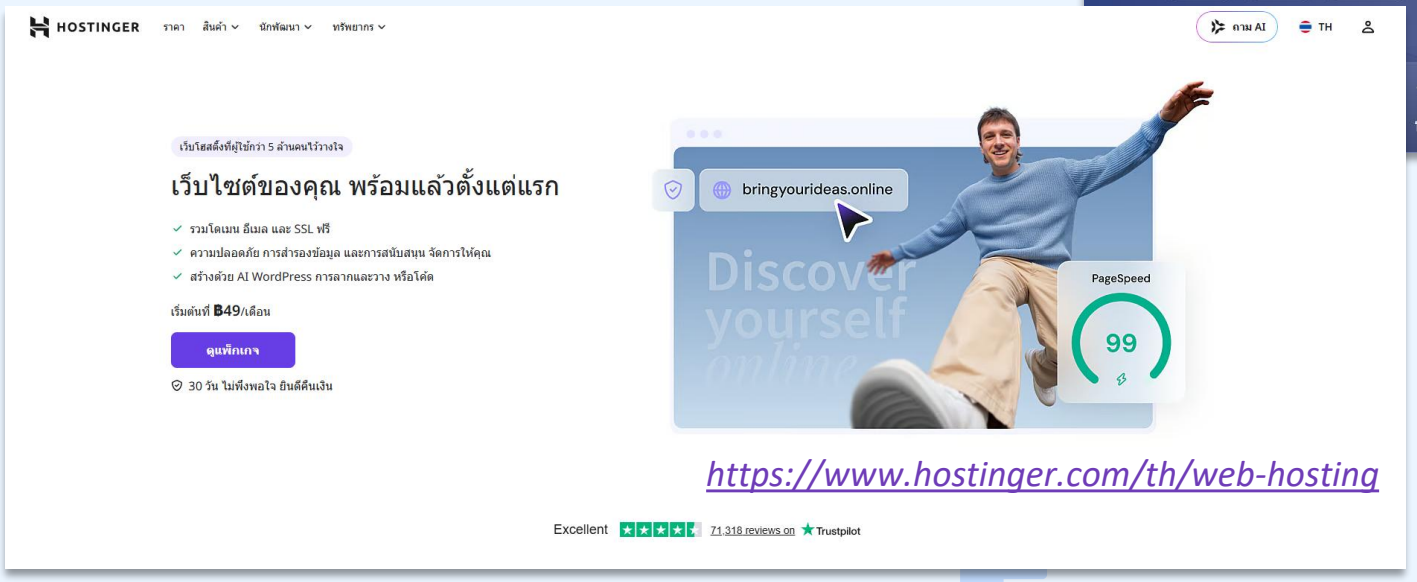

1. The client makes request for service to server
2. The server responds to that request

# Web Hosting

**Web hosting** is a service that provides server storage and infrastructure to host website files, code, media, and databases. Ensuring your site remains online 24/7 and globally accessible via a domain name.




https://www.thnic.co.th/



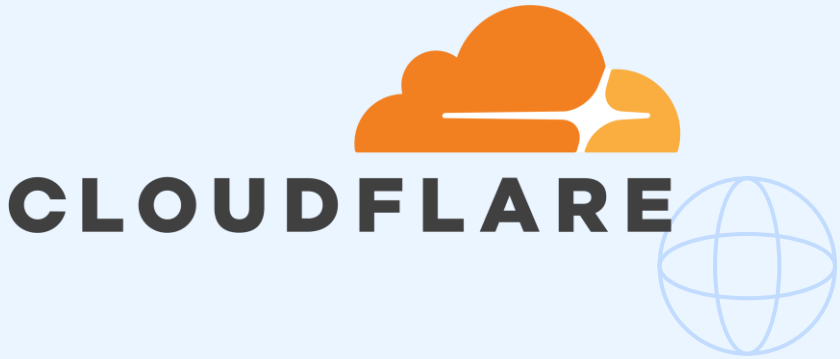

https://www.hostinger.com/th/web-hosting

# Web Hosting (Continue)











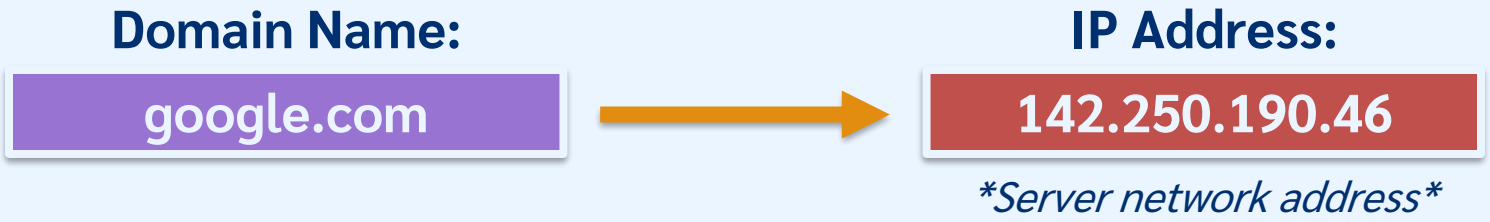




Google Cloud Platform

# What is Domain Name?

A **Domain Name** serves as a **human-friendly alias** that replaces complex numerical **IP Addresses** (e.g., 142.250.190.46), functioning through the **DNS (Domain Name System)** like a global phonebook to automatically translate readable website names (e.g., google.com) into machine-readable network addresses so users never have to memorize raw numbers to browse the web.



# How to choose a Domain Name?

Choosing a domain name (website name) is like choosing a sign for your online store. It should be memorable, reflect your business, and help customers find you easily.

1. **Instant Memorability:** The “Radio Test”, if someone hears the name once, they should be able to type it without asking how to spell it.
2. **Concise & Punchy:** Aim for 1–2 words, target 6–14 characters (e.g., *Tesla, Apple*) for easy typing and memorability.
3. **Easy to Spell & Pronounce:** Avoid ambiguous spellings so users can type it accurately from word-of-mouth.
4. **Avoid Hyphens (-) & Numbers:** Eliminates input errors and user confusion (e.g., mixing up 1 and one).
5. **Brandable over Keywords:** Choose catchy, distinctive names instead of stuffed generic keywords.
6. **Trademark Safety:** Steer clear of existing brand names (e.g., toyota-secondhand) to dodge lawsuits and trust issues.

# Choosing the Right TLD (Top-Level Domain)

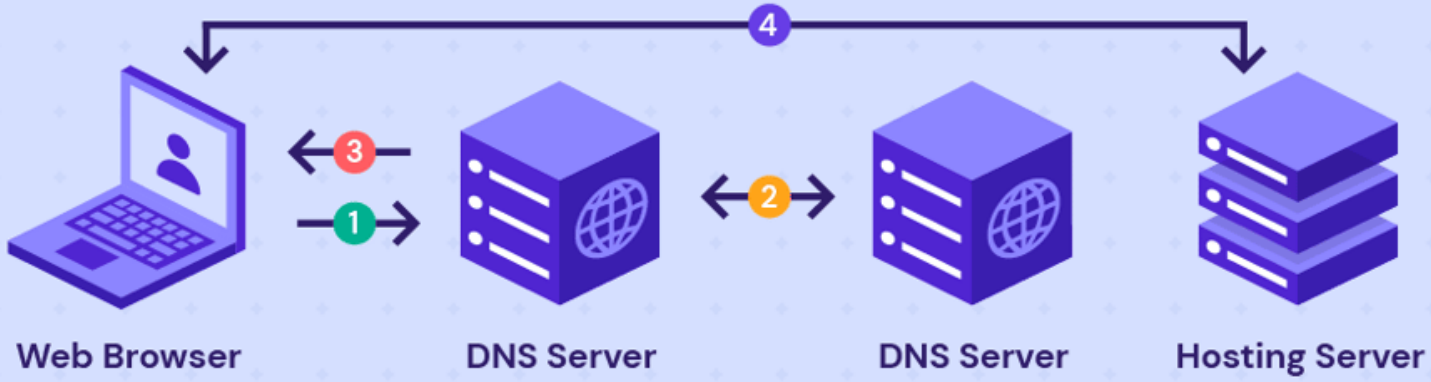
- **.com:** The global standard; delivers the highest familiarity, credibility, and authority.
- **.th / .co.th:** Best for officially registered businesses targeting local markets within Thailand.
- **.dev / .app:** Built for web applications, software products, and developer tools (enforces HTTPS by default).
- **.io / .ai / .tech:** Highly popular among tech startups, SaaS platforms, and AI
- **.org:** The gold standard for non-profits, open-source projects, public initiatives, and community organizations.
- **.net:** A classic legacy alternative to .com, originally intended for network infrastructure and internet service providers.
- **.edu / .ac.th:** Strictly reserved for accredited educational institutions, universities, and academic research bodies.
- **.gov / .go.th:** Restricted exclusively to official government entities, agencies, and public administration portals.
- **.me / .site / .space:** Ideal for personal portfolios, resumes, individual creators, and freelance showcases.
- **.store / .shop:** Clear, dedicated extensions specifically designed for e-commerce platforms and direct-to-consumer online stores.

# *Domain and Web Hosting*

HOSTINGER

Three. Two. Online

## How Do Domain Names Work?




1. When a user enters a domain name in a browser, the server will search through a global server network that makes up the Domain Name System (DNS).
2. The request is made for one DNS server and if the address is not found there, it moves to the other.
3. The DNS server that has information about the IP address will return it to the browser.
4. The browser will request data about the site from the domain's hosting server. Once the hosting server sends the data back, the web browser will convert it into a web page.

# Classroom Rules

1. เข้าเรียนตรงเวลา
2. งานทุกงานโดย Default ห้ามสงงานจาก AI (ยกเว้นงานที่ระบุให้สามารถใช้ได้)
   - สามารถใช้ AI เพื่อช่วยให้คำแนะนำ หรือแก้ไขปัญหาได้
   - หากคะแนนที่ได้รับในแต่ละงาน ไม่ถูกต้องให้ยื่นอุทธรณ์กับ อ.ผู้สอนเสมอ
3. ห้ามคัดลอกงานของเพื่อนส่ง
4. ห้ามส่งเลยกำหนด หากส่งเลยกำหนดส่งจะหักคะแนนงาน วันละ 20%
5. ส่งงานผ่านช่องทาง และรูปแบบที่กำหนดเท่านั้น
6. หากทำผิดกฎใด ๆ อาจารย์ผู้สอนมีสิทธิ์ตัดคะแนน ไม่ตรวจรับงาน และคิดเป็น 0 คะแนน

# *Thank you!*